Effects of Fine Dust on Subjective Well-being of Adolescents: Evidence from South Korea

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Introduction

Motivation and research questions

- Particulate air pollution (fine dust) has a negative impact on human activity and health, with a more deleterious impact on children and the elderly (Jia and Ku, 2019; Rückerl et al., 2011)
  - While there are other indicators (i.e., nitrogen dioxide, sulfur dioxide, carbon monoxide, and ozone), PM poses the most harmful threat to human health (World Health Organization , WHO) and it is capable of easily entering indoor spaces and the human body due to its microscopic size.
- This paper studies the impact of fine dust on the subjective well-being of adolescents.
  - Does a higher level of fine dust negatively impact adolescents' subjective well-being?
  - Does bad air quality have different consequences for different socioeconomic and demographic groups such as gender and parents' education level?

Previous Literature

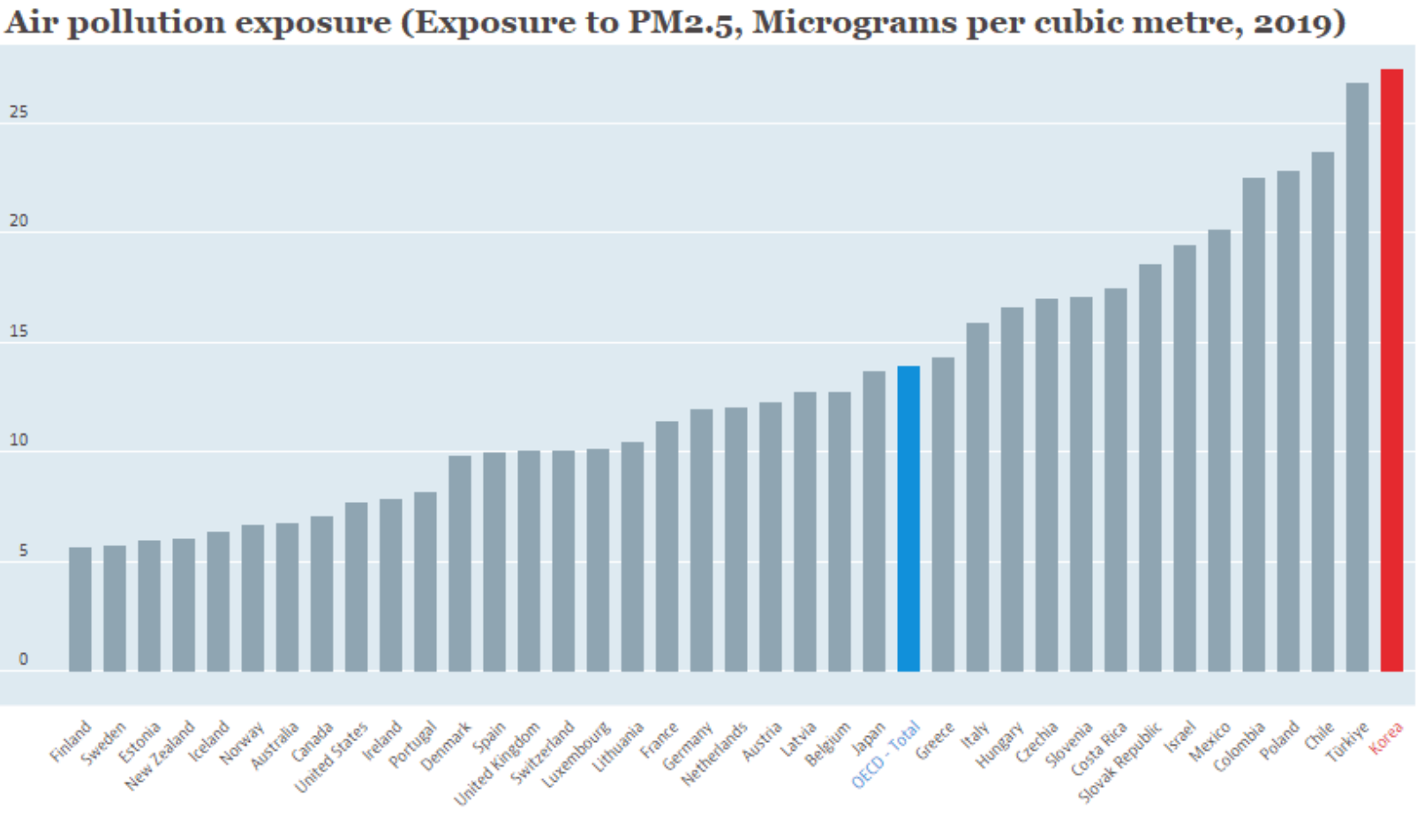
- Particulate air pollution and physical health outcomes
  - Associated with respiratory and cardiovascular disease and hospital admissions for these conditions, such as asthma, bronchitis, lung and heart dysfunction, and cancer (Pope III et al., 2009; EPA, 2009; Brook et al., 2010; Hicken et al., 2014)
- Negative health effects of particulate air pollution on children
  - Cross-border pollution spillover from China increases pollution rated mortality rates in South Korea with a more significant effect seen in children under five and elderly people (Jia and Ku, 2019)
  - Particulate air pollution negatively affects academic performance and school absenteeism for children (Lu et al., 2021; Pope III et al., 1992)
- Negative impact of air pollution on labor outcome
  - Higher level of air pollution reduces labor supply measured by absenteeism or productivity levels (Chang et al., 2016; Graff Zivin and Neidell, 2013; Aragon et al., 2017; Hanna and Oliva, 2015)

Literature & Contribution

- One of the first studies that document the impact of fine dust on adolescents' subjective well-being.
  - Lin et al. (2019) use Taiwan data in 40 districts and examine the effect of the concentration level of air pollution on adolescents' happiness
- The findings may provide empirical grounds for early intervention regarding environmental factors.

Korea Background

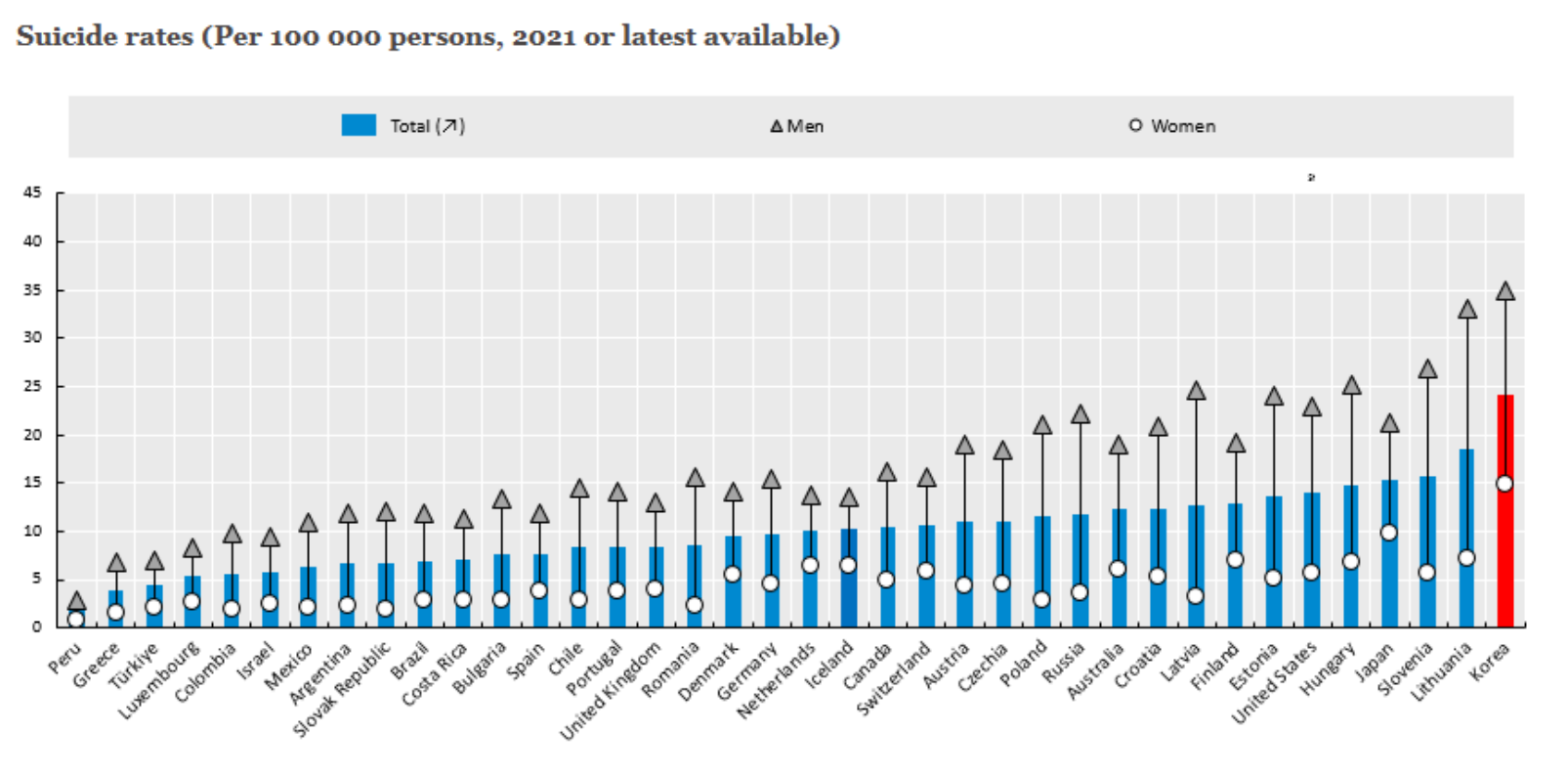
- High concentration of fine dust



Source: OECD (2024), Air pollution exposure (indicator). doi: 10.1787/8d9dcc33-en (Accessed on 31 May 2024)

Korea Background

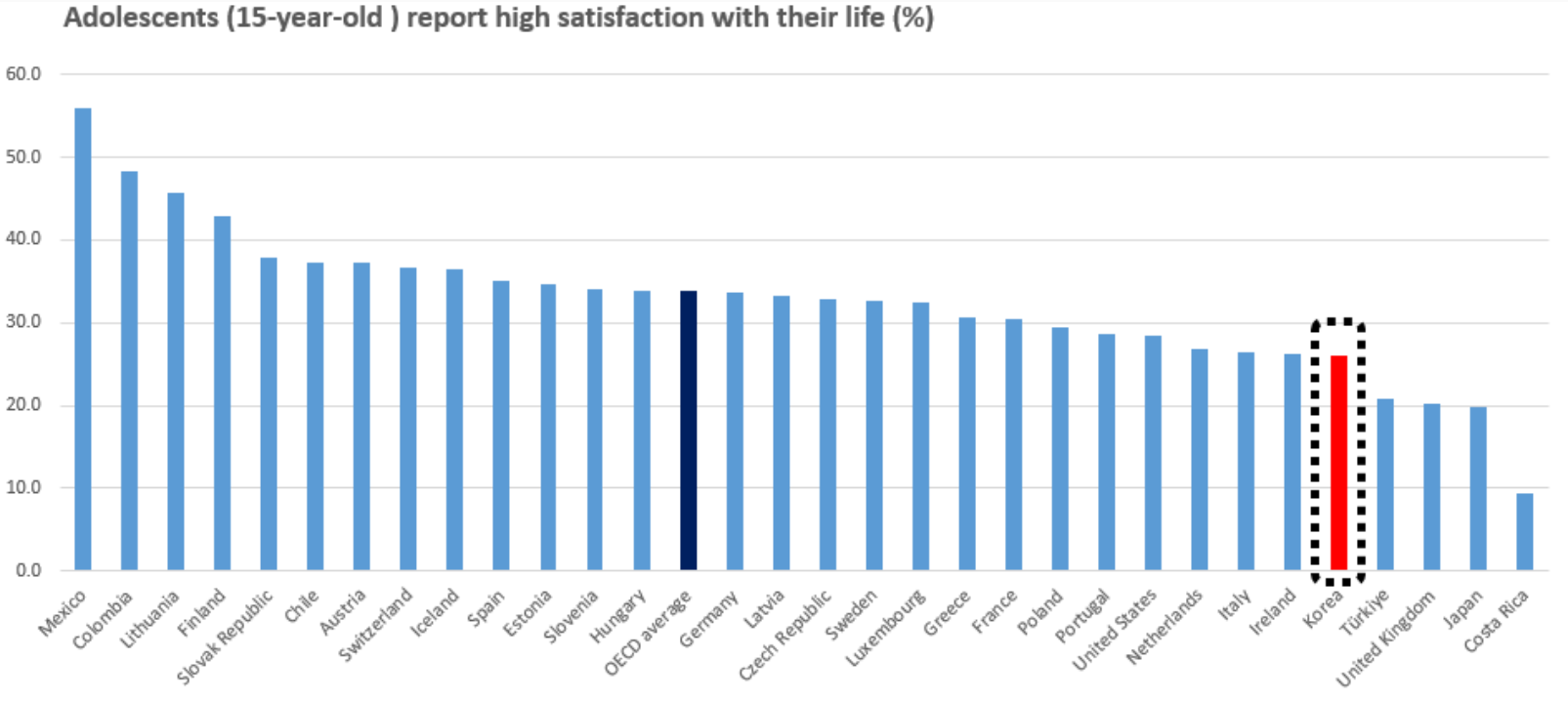
- the highest suicide rates in OECD countries



Source: OECD (2024), Suicide rates (indicator). doi: 10.1787/a82f3459-en (Accessed on 31 May 2024)

Korea Background

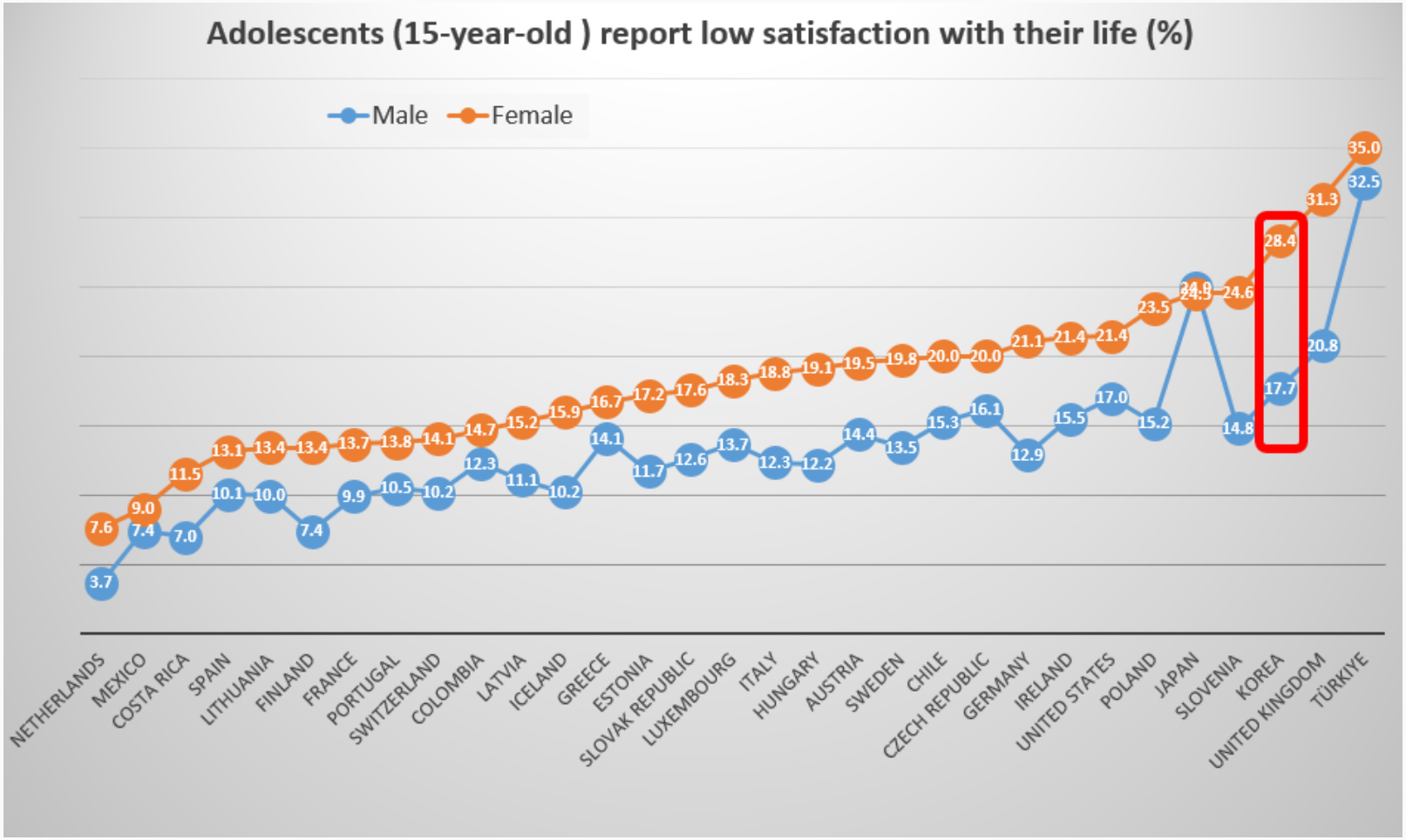
- One of the lowest level of life satisfaction among adolescents in OECD countries



Source: OECD Child Well-Being Data Portal based on 2018 Database

Korea Background

- Adolescents life satisfaction by gender

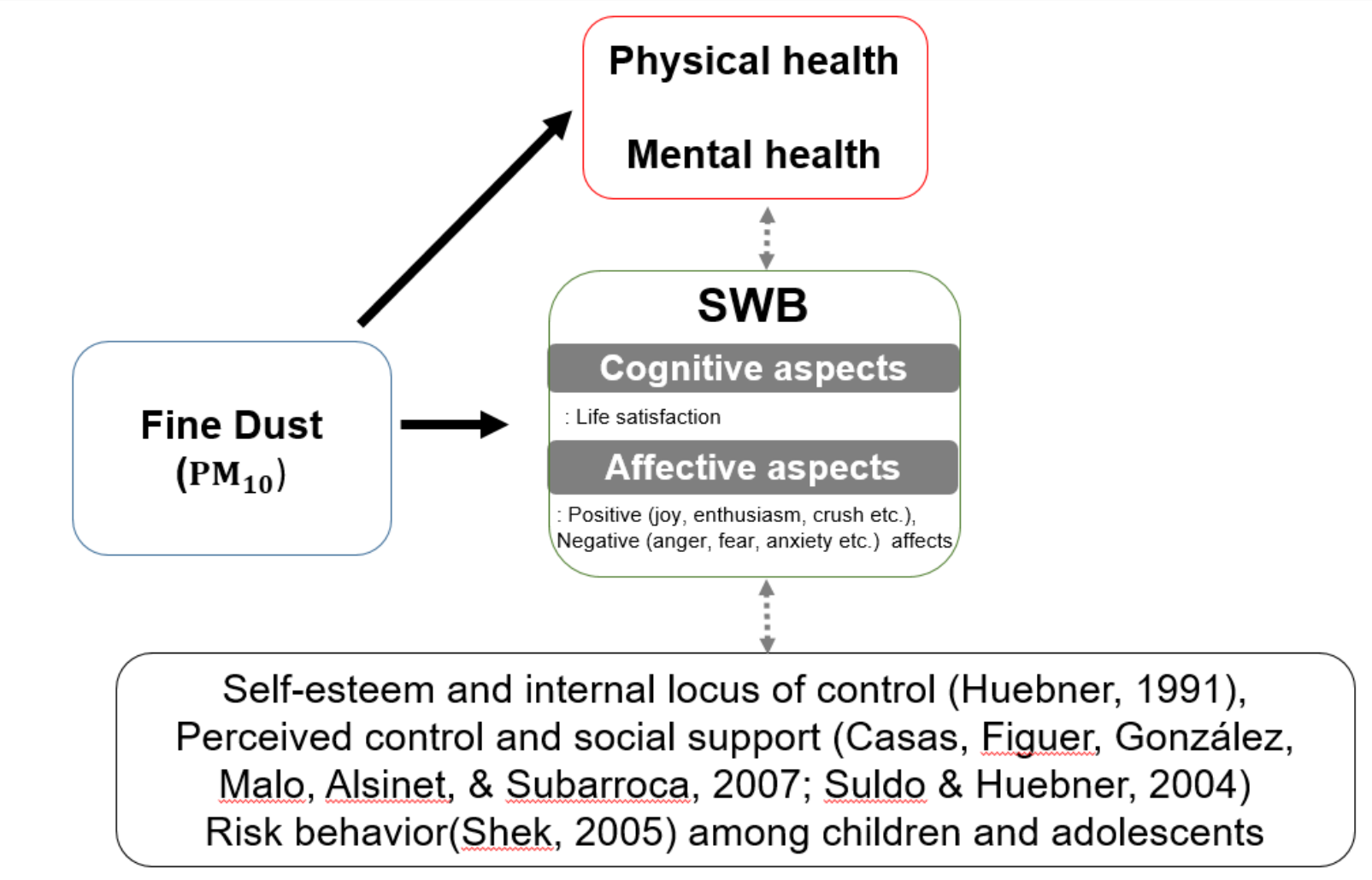


Source: OECD Child Well-Being Data Portal based on 2018 Database

Research Framework

- Subjective well-being (SWB) is defined as 'a person's cognitive and affective evaluations of his or her life' (Diener et al., 2003)
- Our paper examines multi-dimension of SWB: cognitive aspects (life satisfaction), affective aspects (happiness, depression, anxiety).
- SWB is known to be associated with better mental health outcomes (reduces depressive symptoms (Bolier et al., 2013)) and physical health outcomes (disease progression and recovery).

Conceptual Framework



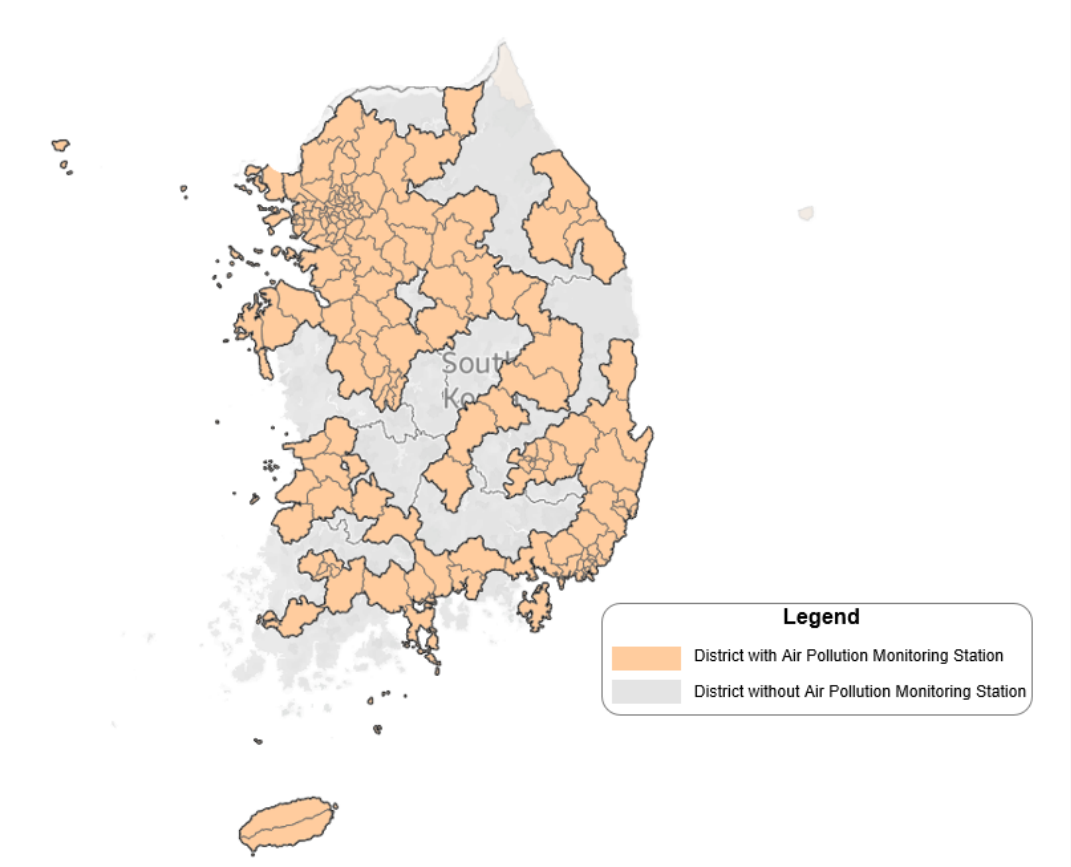
Data & Empirical Strategy

Data

- The Korean Children and Youth Panel Survey in 2010-2016 (KCYPS 2010)
  - Contain rich information including student's SWB and its determinants
- Air pollutant data on the *Airkorea* website (www.airkorea.or.kr)
  - Student's residential address matched with PM_{10} level on a district (Gu) level
- Weather data by the Korea Meteorological Administration
  - AWS (Automatic weather system) data
  - Monthly average of weather variables (temperature, wind speed, rainfall) during survey period
  - 510 measurement stations throughout country

Data

- Approximately 1 station every 4–5km (within Seoul, where about 50% of national population lives in)
- Monitors located in accordance to population density



Subjective well-being variables

- Life satisfaction: cognitive aspect of SWB
  - The item asked students, "I think I'm happy with my life" on a 4-point Likert scale (1=Not at all, 2=No, 3=Yes, 4=Very much)
  - Compared to affective components, life satisfaction is known as the most stable cognitive component of SWB and is analyzed more frequently than the affective components (Suldo et al., 2006)
- While affectivity tends to be stable over time and across situations, several studies indicate that air pollution could negatively affect affectivity (Welsch, 2006; Cuñado and Pérez De Gracia, 2013)
- Positive affect using happiness with the item 'I enjoy living' on a 4-point Likert scale and using a standardized index
- Negative affect using questions related to anxiety and depression
  - This is created by factor analysis using a set of related questions

Empirical Issues

- Inaccurate assignment of pollution levels due to mobility bias of individuals
  - Middle and high school students in Korea are randomly allocated to schools within their residential district due to the 'equalization policy,' which makes Korean students much less mobile
- Nonrandom assignment of pollution levels due to preferences over residential areas depending on employment opportunities, environmental quality etc.
  - Control for individual fixed effects
  - Some time-varying individual characteristics of income, parent's education level included
- Avoidance behavior of pollution (Graff Zivin and Neidell, 2013)
  - Market based: eg purchasing air filters → Heterogeneous effect by socio-economic status
  - Non-market based: eg staying inside → Look at time use

Determinants of SWB in adolescents

- Social support (Siedlecki et al., 2014)
  - Peer, familial support important to SWB: How often does a family member help you with your work for school?", "How often do you talk with your parents about school", "How much do your parents value high scholastic attainment?"
  - Spending time talking to one's children about school important correlate of life satisfaction
- Academic achievement (Crede et al., 2015; Suldo et al., 2006)
  - The research findings on the relationship between academic achievement and life satisfaction in adolescents are mixed

Determinants of SWB in adolescents

- Family structure (Yárnoz-Yaben and Garmendia, 2016)
  - Divorce negative impact on SWB (Yárnoz-Yaben and Garmendia, 2016)
  - Adolescents from an intact family with a good intra-parental relationship and where grandparents were co-resident experienced better SWB (Lin et al., 2019)
- Parenting style (Garcia et al., 2019; Aquilino and Supple, 2001)
  - Parental style with higher parental warmth and lower parental strictness (i.e., indulgent) were found to have higher SWB (Garcia et al., 2019)

Empirical Strategy

$$y_{ijt} = \alpha + \beta_1 PM_{10jt} + \beta_2 PM^2{}_{10jt} + W'_{jt}\gamma + X'_{it}\delta + \sigma_i + \tau_t + \epsilon_{ijt} (1)$$

- y_{ijt}: Outcome variable for individual i in district j during survey year t
- PM_{10jt}: Pollution level (PM_{10}) in district j during survey year t (divided by 10)
- W'_{jt}: Weather controls (average temperature, wind speed, rainfall)
- X'_{it}: Individual characteristics (age, parental education, income, variables of peer and parents relation)
  - Level of victimization individuals experienced and social support from peers, level of supportive parenting and abuse experienced from their parents
- σ_i, τ_t: Individual fixed effect, Year fixed effect

Results & Conclusion

Summary statistics

Variable	Description	Obs.	Mean	Std. Dev.	Min	Max
Panel A. Subjective Well-being Variables						
Life satisfaction (LS)	I think my life is happy	14,099	3.11	0.71	1	4
Positive affect (Happiness)	I enjoy living my life	14,098	3.17	0.67	1	4
Positive affect (Standardized happiness)	I enjoy living my life	14,098	0.00	1.00	-3.23	1.24
Negative affect (Depression)	FA of depression index	11,524	0.00	1.00	-1.72	3.52
Negative affect (anxiety)	FA of anxiety index	11,524	0.00	1.00	-1.41	3.81
Panel B. Time Variables						
Sleeping time (Hour)	Average Sleeping times in a day	15,534	8.35	1.47	2.00	14.17
Private-tutoring time (Hour)	Average private-tutoring times in a day	8,568	1.06	1.27	0.00	10.00
Study-time (Hour)	Average study (homework) times in a day	15,291	2.27	1.92	0.00	17.00
Reading-time (Hour)	Average reading books (not textbooks) times in a day	15,510	0.53	0.70	0.00	7.00
Digital-entertaining time (Hour)	Average Playing on a computer or console, and watching TV, DVD times in a day	15,469	3.36	1.93	0.00	17.00
Friend-time (Hour)	The average time of day to play with friends	15,491	1.46	1.36	0.00	10.00
Panel C. Air Pollution Variables						
PM_{10} ($\mu g/m^3$)	Mean of PM_{10} (Oct.-Dec.)	15,717	46.54	8.63	22.25	100.87
SO_2 (ppm)	Mean of SO_2 (Oct.-Dec.)	15,717	0.005	0.002	0.001	0.01
NO_2 (ppm)	Mean of NO_2 (Oct.-Dec.)	15,717	0.03	0.01	0.004	0.06
CO (ppm)	Mean of CO (Oct.-Dec.)	15,717	0.57	0.16	0.21	1.18
O_3 (ppm)	Mean ofO_3 (Oct.-Dec.)	15,717	0.02	0.01	0.01	0.05

Effects of Fine Dust on Life Satisfaction

	Life Satisfaction	Life Satisfaction	Life Satisfaction
	(1)	(2)	(3)
PM_{10} ($\div 10$)	-0.109***	-0.106***	-0.101***
	(0.038)	(0.036)	(0.036)
PM_{10}^2 ($\div 100$)	0.010***	0.010***	0.010***
	(0.003)	(0.003)	(0.003)
Observations	14,099	13,837	13,837
R-squared	0.003	0.003	0.029
Mean	3.108	3.108	3.108
Year, Individual FE	Y	Y	Y
Weather controls	N	Y	Y
Individual controls	N	N	Y

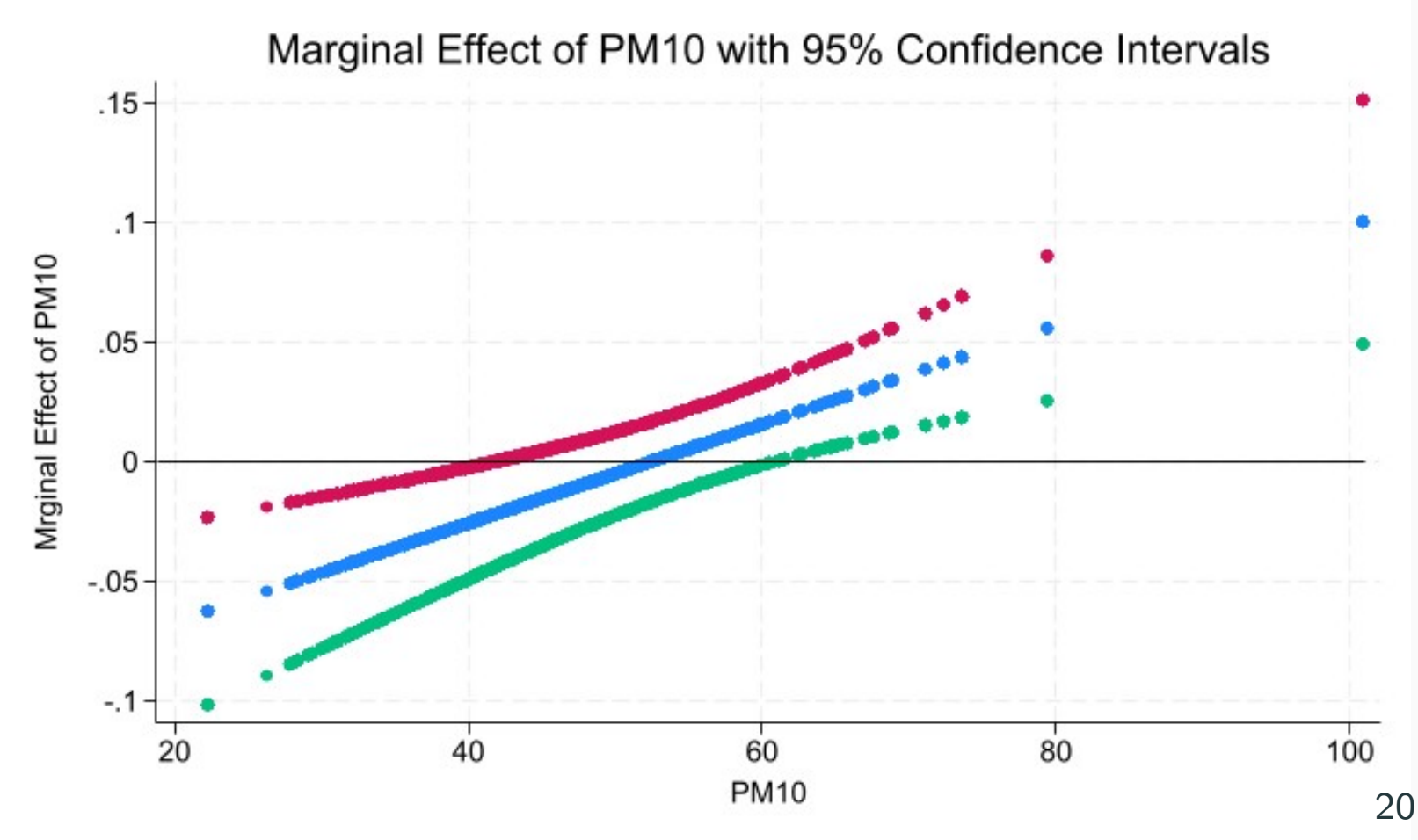
Effects of Fine Dust on Positive and Negative Affects

	Positive affect (Happiness)	Positive affect (Standardized happiness)	Negative affect (Anxiety)	Negative affect (Depression)
	(1)	(2)	(3)	(4)
PM_{10} (÷10)	-0.091**	-0.136**	0.263**	0.111
	(0.043)	(0.064)	(0.104)	(0.086)
PM_{10}^2 (÷100)	0.008*	0.011*	-0.028**	-0.013
	(0.004)	(0.006)	(0.011)	(0.010)
Observations	13,836	13,836	11,300	11,300
R-squared	0.033	0.033	0.103	0.083
Mean	3.168	-0.000	0.003	-0.000
Year, Individual FE	Y	Y	Y	Y
Weather controls	Y	Y	Y	Y
Individual controls	Y	Y	Y	Y

Marginal effect

- There is non-linear relationship between PM_{10} concentration and subjective well-benig outcomes
  - Include squared term of PM_{10} in the fixed effect model

$$dY/dPM_{10} = \beta_1 + 2\beta_2 PM_{10} \tag{2}$$



Potential channels

	Sleeping time	Private-tutoring time	Study-time	Reading-time	Digital-entertaining time	Friend-time
PM_{10} (÷10)	-0.273**	0.078	-0.177	-0.078	-0.001	0.088
	(0.124)	(0.147)	(0.179)	(0.048)	(0.148)	(0.111)
PM_{10}^2 (÷100)	0.025**	-0.005	0.022	0.007*	-0.006	-0.005
	(0.011)	(0.011)	(0.014)	(0.004)	(0.012)	(0.009)
Observations	15,237	8,397	15,006	15,214	15,173	15,193
R-squared	0.359	0.050	0.064	0.027	0.132	0.010
Mean	8.362	1.038	2.254	0.525	3.377	1.468
Year, Individual FE	Y	Y	Y	Y	Y	Y
Weather controls	Y	Y	Y	Y	Y	Y
Individual controls	Y	Y	Y	Y	Y	Y

- When PM_{10} increases by 10 units, sleep duration significantly decreases by approximately 20 minutes
  - Exposure to ambient air pollutants has detrimental effects on sleep quality and sleep duration (Cao et al., 2021; Chen et al., 2019; Liu et al., 2020) and affects sleep-disordered breathing (wheezing and snoring) in children (Sánchez et al., 2019; Lawrence et al., 2018)

Heterogeneity Effects of Fine Dust by gender and SES

	Life Satisfaction		Life Satisfaction	
	Male	Female	Low SES	High SES
PM_{10} ($\div 10$)	-0.126***	-0.078	-0.089*	-0.120
	(0.045)	(0.056)	(0.049)	(0.084)
PM_{10}^2 ($\div 100$)	0.013***	0.006	0.009**	0.012
	(0.004)	(0.004)	(0.004)	(0.009)
Observations	7,441	6,396	5,869	7,968
R-squared	0.029	0.036	0.028	0.032
Mean	3.155	3.053	3.064	3.140
Year, Individual FE	Y	Y	Y	Y
Weather controls	Y	Y	Y	Y
Individual controls	Y	Y	Y	Y

Potential channels (Gender)

	Sleeping time		Private-tutoring time		Study-time	
	Male	Female	Male	Female	Male	Female
PM_{10} (÷10)	-0.487***	0.041	-0.025	0.262	-0.143	-0.261
	(0.140)	(0.110)	(0.136)	(0.209)	(0.204)	(0.292)
PM_{10}^2 (÷100)	0.045***	-0.004	0.001	-0.017	0.018	0.029
	(0.013)	(0.009)	(0.011)	(0.015)	(0.016)	(0.023)
Observations	8,229	7,008	4,660	3,737	8,096	6,910
R-squared	0.373	0.349	0.069	0.039	0.055	0.088
Mean	8.391	8.327	1.047	1.027	2.035	2.511
Year, Individual FE	Y	Y	Y	Y	Y	Y
Weather controls	Y	Y	Y	Y	Y	Y
Individual controls	Y	Y	Y	Y	Y	Y

Potential channels (Gender)

	Reading-time		Digital-entertaining time		Friend-time	
	Male	Female	Male	Female	Male	Female
PM_{10} (÷10)	-0.104	-0.050	-0.123	0.178	-0.076	0.336**
	(0.067)	(0.067)	(0.145)	(0.185)	(0.134)	(0.163)
PM_{10}^2 (÷100)	0.010	0.004	0.004	-0.021	0.010	-0.027**
	(0.006)	(0.005)	(0.012)	(0.014)	(0.012)	(0.013)
Observations	8,204	7,010	8,192	6,981	8,198	6,995
R-squared	0.026	.031	0.102	0.178	0.014	0.013
Mean	0.522	0.528	3.552	3.172	1.517	1.411
Year, Individual FE	Y	Y	Y	Y	Y	Y
Weather controls	Y	Y	Y	Y	Y	Y
Individual controls	Y	Y	Y	Y	Y	Y

Potential channels (SES)

	Sleeping time		Private-tutoring time		Study-time	
	Low SES	High SES	Low SES	High SES	Low SES	High SES
PM_{10} (÷10)	-0.327*	-0.268**	-0.027	0.138	-0.258	-0.072
	(0.181)	(0.116)	(0.148)	(0.186)	(0.161)	(0.293)
PM_{10}^2 (÷100)	0.029**	0.027**	0.004	-0.009	0.030**	0.010
	(0.014)	(0.011)	(0.011)	(0.016)	(0.012)	(0.026)
Observations	6,496	8,741	3,831	4,566	6,449	8,557
R-squared	0.309	0.395	0.078	0.036	0.046	0.085
Mean	8.466	8.284	0.800	1.238	1.813	2.586
Year, Individual FE	Y	Y	Y	Y	Y	Y
Weather controls	Y	Y	Y	Y	Y	Y
Individual controls	Y	Y	Y	Y	Y	Y

Potential channels (SES)

	Reading-time		Digital-entertaining time		Friend-time	
	Low SES	High SES	Low SES	High SES	Low SES	High SES
PM_{10} (÷10)	-0.104	-0.050	-0.123	0.178	-0.076	0.336**
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Year, Individual FE	Y	Y	Y	Y	Y	Y
Weather controls	Y	Y	Y	Y	Y	Y
Individual controls	Y	Y	Y	Y	Y	Y

Robustness check

- Our results are robust to
  - Including all air pollutants
  - Various samples
    - (1) Area measurement issue: small area (City, Si (in Korean)), large area (State, Do (in Korean))
    - (2) Mobility bias issue: Only non-movers in the survey period
  - Different analysis: using ordered probit model

Robustness Checks (I): All air pollutants

	(1)	(2)	(3)	(4)	(5)
PM_{10} (÷10)	-0.104***	-0.102***	-0.112***	-0.104***	-0.114***
	(0.036)	(0.036)	(0.035)	(0.036)	(0.035)
PM_{10}^2 (÷100)	0.010***	0.010***	0.010***	0.010***	0.011***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
SO_2	8.261				5.730
	(10.075)				(9.852)
NO_2		0.268			-0.380
		(2.124)			(2.067)
CO			0.161**		0.155*
			(0.077)		(0.080)
O_3				2.131	1.957
				(2.838)	(2.904)
Observations	13,837	13,837	13,837	13,837	13,837
R-squared	0.029	0.029	0.029	0.029	0.029
Year, Individual FE	Y	Y	Y	Y	Y
Weather controls	Y	Y	Y	Y	Y
Individual controls	Y	Y	Y	Y	Y

Robustness Checks (II): by Area size, Non-movers, Ordered probit

	(1) Small area	(2) Large area	(3) Non-movers	(4) Ordered probit
PM_{10} ($\div 10$)	-0.246***	-0.078*	-0.103***	-0.202***
	(0.090)	(0.041)	(0.037)	(0.074)
PM_{10}^2 ($\div 100$)	0.025**	0.008**	0.009**	0.017***
	(0.010)	(0.003)	(0.003)	(0.007)
Observations	5,185	8,652	13,198	13,198
R-squared	0.037	0.028	0.029	
Mean	3.146	3.085	3.110	
Year, Individual FE	Y	Y	Y	Y
Weather controls	Y	Y	Y	Y
Individual controls	Y	Y	Y	Y

Conclusion

- This paper examines the impact of fine dust on adolescents' subjective well-being.
  - An increase in the level of PM_{10} negatively affects the subjective well-being of adolescents.
  - The estimated effect size is nearly identical to the estimated effect size of the relationship air pollution and adolescents' happiness (Lin et al., 2019)
- There's a strong non-linear effect of PM_{10} where the negative effect of PM_{10} becomes weaker with the higher level of concentration; possibly because at higher concentration levels weather forecasts and government alerts warn citizens to watch out for fine dust (knowing may reduce perceived risk).

Conclusion

- We examine the heterogeneity Effects of Fine Dust on SWB by gender and SES.
  - Males are slightly more affected by PM_{10}
    - If PM_{10} levels increase by 10 units, sleep duration of male students statistically significantly decreased by 0.5 hours; female students increased by 0.2 hours in private tutoring time, 0.2 hours in digital entertainment time, and 0.3 hours in friend time
  - The effect is more concentrated among students with lower parental education (SES)
    - For both groups, sleep duration decreased significantly by approximately 0.3 hours
    - In high SES, private tutoring time increases by 0.14 hours, and digital entertaining time increases by 0.2 hours, but it is not statistically significant; friend time statistically significantly increases by 0.3 hours

Thank you!

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